

Question1

A user is playing a game of “**Transformers**” in a computer. The user has entered into a scenario where Megatron, Prime and Bumblebee are in a battle. The features of the battle are given below.

- Megatron, Prime and Bumblebee are three robots and they have unique processes as indicated below.
 - M1 and M2 processes belong to Megatron.
 - P1 and P2 processes belong to Prime.
 - B1 and B2 processes belong to Bumblebee.
- At the beginning all the robots have a health of 100 units
- Each process is considered as a “Hit” and after completion of each process its considered as a hit to opponent.
- After completion of each process (Hit) belongs to Megatron and Prime will result a 50 units of health loss on the opponent.
- After completion of each process (Hit) belongs to Bumblebee will result a 25 units of health loss on the opponent.
- Prime and Bumblebee are in the same side and they won’t attack each other but they will try to attack and kill Megatron.
- Megatron on the other hand will try to attack Prime and Bumblebee. Megatron will first check whether bumblebee is alive and hit or go for a hit on Prime.
- If the health of a robot goes to 0 units or below it implies the robot is killed

Suppose that the following processes arrive for execution at the times indicated. Each process will run for the duration given. In answering the questions, use non preemptive scheduling and base all your decisions on the information you have at the time the decision is made. Length of the CPU burst and also the arrival time to the ready queue are given in milliseconds.

Process	Arrival Time	Burst Time
M1	2	12
M2	3	15
P1	0	10
P2	4	13
B1	5	5
B2	1	7

- (i) Draw a Gantt chart to illustrate the execution of the above processes using each algorithm mentioned below and calculate the average waiting time and average turnaround time.
 - (a) First Come First Served (FCFS),
 - (b) Shortest Job First (SJF)
 - (c) Round Robin with a time quantum of 6 milliseconds.
- (ii) After completion of each algorithm mentioned in section (a) for the set of processes mentioned above, answer the following questions:
 - (a) Indicate the health of each robot.
 - (b) Indicate the instances where Megatron gets killed before his first Hit.
 - (c) Explain why Prime is alive in all situations.

Question2

In an operating system there is a shared variable “x”.

The relevant functions for “addnum” and “addsum” are given below :

```
int addnum(int x)
{
    int t, ad = 0, remainder;
    t = x;
    while (t > 1)
    {
        remainder = t % 10;
        ad      = ad + remainder;
        t      = t / 10;
    }
    x = ad;
    return x;
}
```

```
int addsum(int x)
{
    int n, sum = 1, c;
    n=x;
    for (c = 1; c <= n; c++)
    {
        sum = sum + sum * c;
    }
    x = sum;
    return x;
}
```

Answer the following questions using the above description.

- (a) Explain the race condition using the above two functions.
- (b) Identify the critical section for each function and write the necessary code segments to demonstrate solutions for critical region using:
 - a. Swap.
 - b. Semaphores.
- (c) Show how mutual exclusion and bounded waiting is preserved for each solution mentioned in question (ii).